

Lecture 1

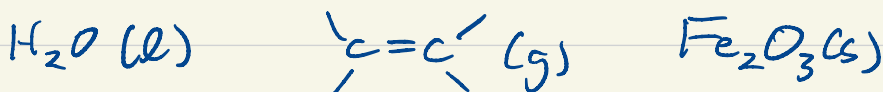
Most important fundamental observation about nature?

Matter is made of discrete & infinitesimal particles, in constant motion.

- Feynman

Concept anticipated by ancient Greek & Indian philosophers

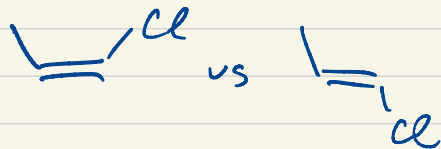
Distinct chemical substances



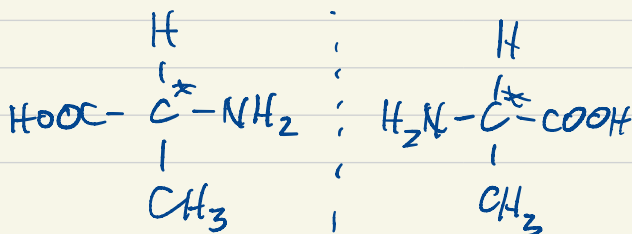
often appearing in mixtures

Distinct can depend on context

1-chloropropene
geometric isomers



alanine
stereoisomers



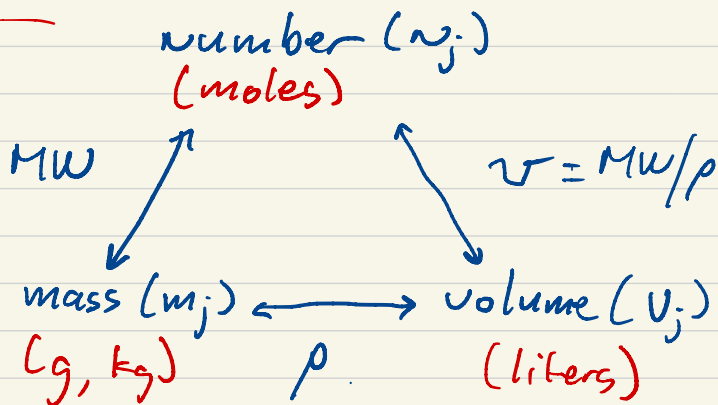
water
isotopomers

H_2O vs D_2O

alumina
polymorphs

$\alpha-Al_2O_3$ vs θ vs γ

Amounts

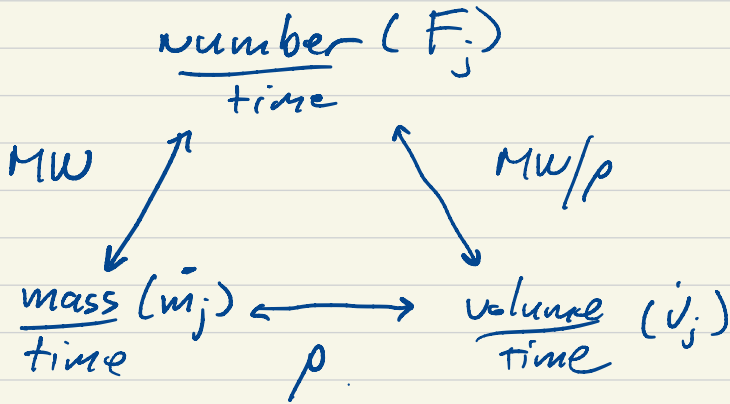
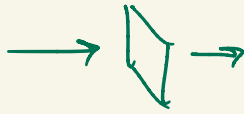


ρ , density
+
 MW/ρ , molar volume

} s, l weak fcn of T, ρ
s strong " " "

$$v = \frac{MW}{\rho} \approx \frac{RT}{p} \cdot Z$$

Flows



Composition (activities)

key connection to equilibrium & to rates

Normalize amts/flow to any extensive quantity

$$n_{\text{tot}} = \sum_j n_j$$

$$m_{\text{tot}} = \sum_j m_j$$

$$V_{\text{tot}} \neq \sum_j \bar{V}_j \quad \text{right? Oh no!}$$

• mole fractions

$$x_j, y_j = \frac{n_j}{n_{\text{tot}}} = \frac{F_j}{F_{\text{tot}}}$$

- concentrations

$$C_j = \frac{N_j}{V_{\text{tot}}} = \frac{F_j}{\dot{V}_{\text{tot}}}$$

- molality $\frac{n_j}{m_{\text{tot}}}$

- partial pressure

$$P_j = y_j \cdot P$$

Example (trivial)

Air is 80% N_2 / 20% O_2 (what are these units?)

Mass flow controller delivers 1 g/s dry air to reactor.

F_{N_2} ? F_{O_2} ? $\dot{V}$? what would you need to know?

$\dot{m} = 1 \text{ g/s}$

$y_{O_2} = 0.2$

$y_{N_2} = 0.8$



298 K

1 bar

Draw picture!!

$$\dot{m} = 1.0 \text{ g air/s}$$

$$MW_{\text{air}} = 0.8 \frac{\text{mol N}_2}{\text{mol air}} \cdot \frac{28 \text{ g N}_2}{\text{mol N}_2} + 0.2 \frac{32 \text{ g O}_2}{\text{mol O}_2}$$

$$= 28.8 \text{ g air/mol air}$$

$$F_{\text{N}_2} = 1.0 \frac{\text{g air}}{\text{s}} \cdot \frac{1 \text{ mol air}}{28.8 \text{ g air}} \cdot 0.8 \frac{\text{mol N}_2}{\text{mol air}} = 230 \frac{\text{mol}}{\text{s}}$$

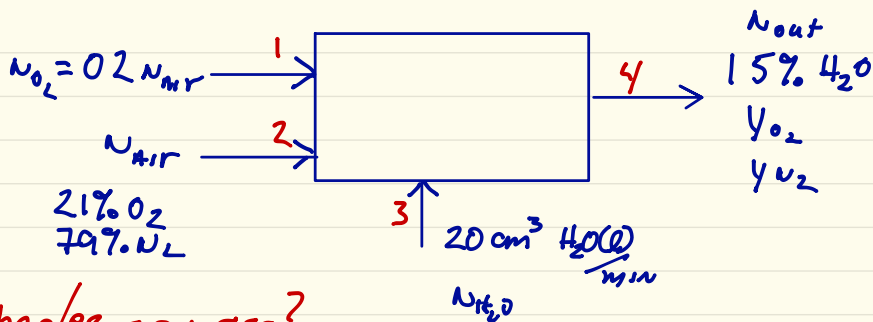
$$F_{\text{O}_2} = \text{ " " " " } 5.8 \frac{\text{mol}}{\text{s}}$$

$$P \dot{V} = F_{\text{tot}} RT$$

$$R = 8.31441 \text{ Pa m}^3/\text{mol K}$$

$$P = 10^5 \text{ Pa}$$

$$\dot{V} = \frac{(F_{\text{N}_2} + F_{\text{O}_2}) R (298 \text{ K})}{P} = 0.714 \text{ m}^3/\text{s}$$



moles or mass?

Water balance

$$n_{H_2O} = \frac{20 \text{ cm}^3}{\text{min}} \cdot \frac{1 \text{ g}}{\text{cm}^3} \cdot \frac{1 \text{ mol}}{18 \text{ g}} = 1.11 \text{ mol}_{H_2O}/\text{min}$$

$$\text{Output} = \text{Input}$$

$$n_{H_2O} = 0.15 n_{out}$$

$$n_{out} = 741 \text{ mol/min}$$

N₂ balance

$$\text{output} = \text{input}$$

$$y_{N_2} \cdot n_{out} = 0.79 n_{Air}$$

O₂ balance

$$y_{O_2} n_{out} = 0.21 n_{Air} + 0.2 n_{Air} = 0.41 n_{Air}$$

unknowns

$$y_{N_2}, y_{O_2}, n_{Air} \quad \text{But} \quad y_{N_2} + y_{O_2} = 0.985$$

3 eqs, 3 unknowns $\rightarrow$ 1 sol'n

Shortcut total mole balance

$$\text{output} = \text{input}$$

$$n_{\text{out}} = n_{\text{H}_2\text{O}} + n_{\text{air}} + 0.2 n_{\text{air}}$$

$$74 \frac{\text{mol}}{\text{min}} = 11 \frac{\text{mol}}{\text{min}} + 1.2 n_{\text{air}}$$

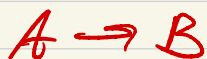
$$n_{\text{air}} = 60.8 \text{ mol/min} \quad n_{\text{O}_2} = 12.1 \text{ mol/min}$$

$$y_{\text{N}_2} = 0.79 \frac{n_{\text{air}}}{n_{\text{out}}} = 0.65 \quad y_{\text{O}_2} = 0.34$$

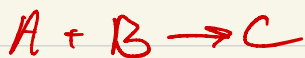
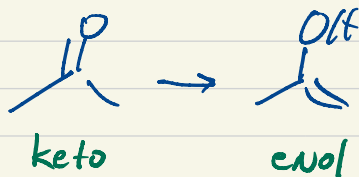
Math is simple Challenge
is to set up correctly
and to choose order of
solution wisely

chemical reaction

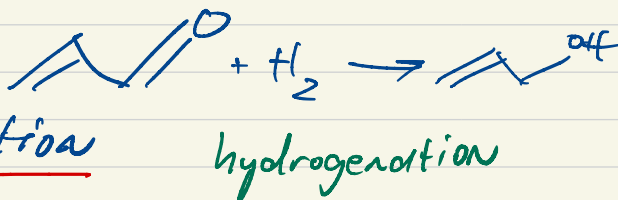
conversion of one substance to another
conserves mass & atoms



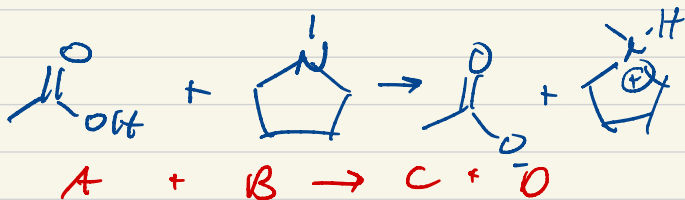
isomerization



condensation



acid-base



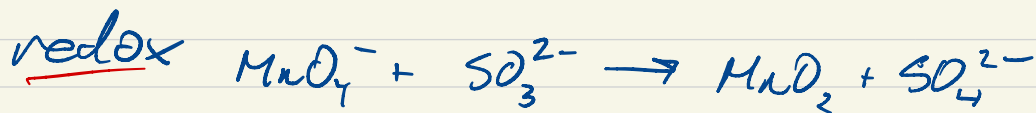
Brønsted acid - Brønsted base
 H^+ donor H^+ acceptor

Lewis acid - Lewis base?

WGS

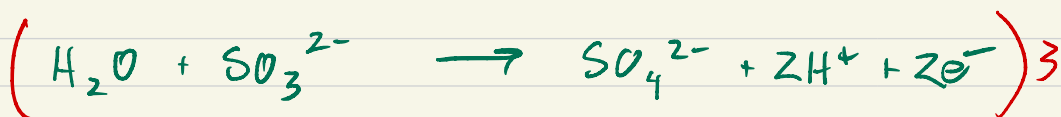


combustion



oxidation states

half-reactions



catalyzed

Distinct stoichiometry

NH_3 combustion



Define stoichiometric coefficient

$$\nu_j = \begin{cases} < 0 & \text{reactant} \\ > 0 & \text{product} \end{cases}$$

$$\sum_j \nu_j A_j = 0 \quad \left\{ \begin{array}{l} \cdot c \text{ can always} \\ \text{scale} \end{array} \right.$$

$$\nu_1 = -4 \quad \nu_2 = -3 \quad \nu_3 = 2 \quad \nu_4 = 6$$

Define advancement / extent ξ

n_{10}	n_{20}	n_{30}	n_{40}
-4ξ	-3ξ	2ξ	6ξ
$n_{10} - 4\xi$	$n_{20} - 3\xi$	$n_{30} + 2\xi$	$n_{40} + 6\xi$

$$n_j = n_{j0} + \nu_j \xi$$

$$F_j = \bar{F}_{j0} + \gamma_j \xi$$

ξ extensive (moles), can be ≥ 0
moles/time

$$\leftarrow \xi < 0 \quad \xi > 0 \rightarrow$$

bounded

$$n_j = n_{j0} + \nu_j \xi \geq 0 \quad \forall j$$

$$\xi \geq -n_{j0}/\nu_j$$

reactants, $\nu_j < 0$

$$\xi \leq -\frac{n_{j0}}{\nu_j}$$

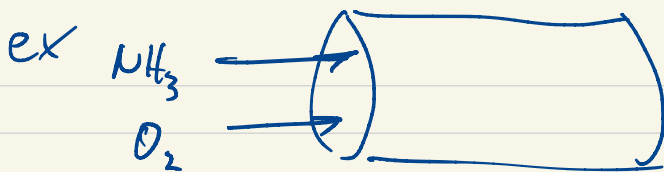
products, $\nu_j > 0$

$$\xi \geq -\frac{n_{j0}}{\nu_j}$$

$$\xi_{\max} = \min_{\text{reactants}} \left[-\frac{n_{j0}}{\nu_j} \right]$$

$$\xi_{\min} = \max_{\text{products}} \left[-\frac{n_{j0}}{\nu_j} \right]$$

$$\xi_{\min} \leq \xi \leq \xi_{\max}$$



$$F_{\text{NH}_3,0} = 0.006 \text{ mol/s}$$

$$F_{\text{O}_2,0} = 0.012 \text{ mol/s}$$

Limiting agent?

$$\text{NH}_3: \frac{-0.006}{-4} = 0.00125 \text{ * limiting}$$

$$\text{O}_2: \frac{-0.012}{-3} = 0.004$$

$$\xi_{\text{max}} = 0.00125 \text{ mol/s}$$

Max O₂ consumption?

$$F_{\text{O}_2} = 0.012 - 3(0.00125) \\ \approx 0.008 \text{ mol/s}$$

Fractional consumption?

$$\frac{F_{\text{O}_2,0} - F_{\text{O}_2}}{F_{\text{O}_2,0}} = \frac{0.012 - 0.008}{0.012} =$$

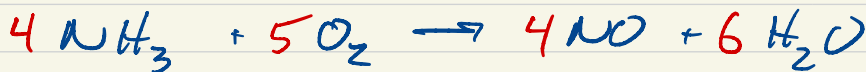
"conversion"

ξ is extensive. Common to
normalize to $\xi_{\max}$

$$X = \xi / \xi_{\max} = - \left(\frac{\gamma_{\text{lim}}}{n_{\text{lim},0}} \right) \xi$$

intensive conversion

Same notation extends to multiple reactions:



$$-4 A_1 - 3 A_2 + 2 A_3 + 6 A_4 = 0$$

$$-4 A_1 - 5 A_2 + 4 A_5 + 6 A_4 = 0$$

$$\underbrace{\begin{pmatrix} -4 & -3 & 2 & 6 & 0 \\ -4 & -5 & 0 & 6 & 4 \end{pmatrix}}_{\gamma} \begin{pmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \\ A_5 \end{pmatrix} = 0$$

Stoichiometric matrix
species vector

$$\gamma \vec{A} = 0 \quad \sum_j \gamma_{ij} A_j = 0 \quad \forall i \text{ rows}$$

Each rxn gets its own advancement ξ_i :

$$n_j = n_{j0} + \sum_i \gamma_{ij} \xi_i$$

$$\vec{n} = \vec{n}_0 + \underline{\gamma} \vec{\xi}$$

$$\text{-or- } F_j = F_{j0} + \sum_i \gamma_{ij} \xi_i$$

Given all flows, can solve for advancements.

"Conversion" ambiguous when multiple rxns are possible.
Avoid, or define carefully.

extent-dependent compositions:

mole fraction

$$x_j, y_j = \frac{n_j}{\sum_j n_j} = \frac{n_{j0} + \nu_j \xi}{\underbrace{\sum_j (n_{j0} + \nu_j \xi)}_{w_{tot}}}$$

concentration

If reactor volume / volumetric flow rate is constant, easy:

$$c_j = n_j / V$$

Often deal w/ constant pressure:

$$\begin{aligned} V &= \sum_j \bar{v}_j(T, P) \cdot n_j \\ &= \sum_j \bar{v}_j (n_{j0} + \nu_j \xi) \end{aligned}$$

If ideal gases, $\bar{v}_j \approx \frac{RT}{P} \quad \forall j$

$$V = \left(\frac{RT}{P} \right) \sum_j (n_{j0} + \nu_j \xi)$$

$$c_j = \frac{n_j}{V} = \left(\frac{RT}{P} \right) y_j$$

Often deal with cases in which there is a diluent in excess

$$w_{\text{tot}} = \sum_j w_j \text{ must include diluent} \\ \approx \text{constant}$$

